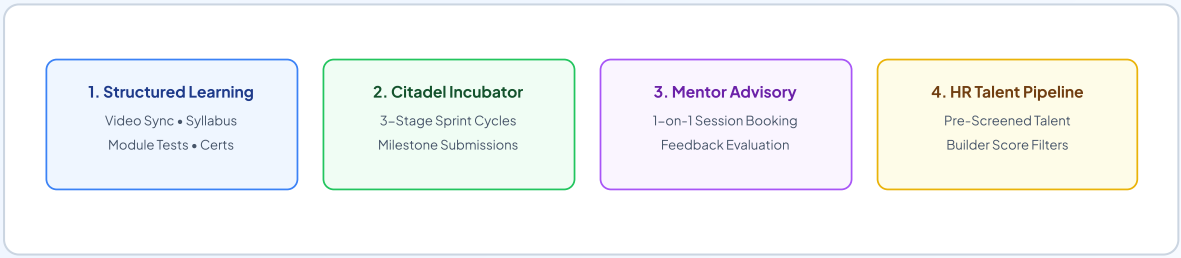
FORGE LOOM LMS

Product Management Document

Product Requirements Document (PRD)

Core strategic product specification detailing market vision, multi-role user journeys, the Citadel Innovation Incubator, algorithmic Builder Scoring, and enterprise HR placement integration.



DOCUMENT PURPOSE
Product Requirements Document (PRD)

PRODUCT MODEL
Dual-Engine LMS & Career Launchpad

TARGET USERS
Students, Mentors, Trainers, Colleges, Recruiters

RELEASE VERSION
v1.0.0 • Production Certified

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1 Product Vision & Strategic Objectives

Traditional Learning Management Systems are passive content repositories. They measure attendance and quiz completion rather than true builder capability. **FORGE Loom LMS** disrupts this paradigm by uniting technical coursework with real-world execution through the **Citadel Innovation Incubator**.

Strategic Value Proposition

Vision

Students do not simply finish video courses; they form cross-functional teams, select real corporate problem statements, execute bi-weekly deliverables across 3 structured sprint cycles, receive qualitative grading from industry mentors, and get shortlisted by corporate HR recruiters based on a verified **Builder Score**.

Core User Personas

The Student Builder

Learner

- **Needs:** Practical skill building, portfolio creation, 1-on-1 industry guidance, and high-paying placement opportunities.
- **Key Jobs:** Track video lesson progress, execute sprint milestones, book mentor sessions, and share verified credential QR codes.

The College Administrator

Institution

- **Needs:** Accreditation compliance, student retention, faculty productivity, and campus placement percentage.
- **Key Jobs:** Monitor department progress, oversee cohort enrollments, inspect faculty logs, and evaluate campus hiring.

2 Core Product Epics & Features

Product Epic	User Stories	Acceptance Criteria
EP-01: Course Engine & Video Progress	As a Student, I want to watch course videos with timestamp tracking so my progress is saved across sessions.	Video resumes at last saved timestamp; lesson completion triggers progress bar recalculation; updates student velocity score.
EP-02: Citadel Incubator & Sprints	As a Student Team, we want to choose an industry problem statement and submit deliverables for Sprints 1, 2, and 3.	Horizontal stepper reflects Sprint status (not started, in progress, submitted, reviewed); Sprint 3 approval unlocks Investor Access banner.
EP-03: 1-on-1 Mentor Advisory	As a Student, I want to book a 30-minute advisory slot with my mentor using an interactive glassmorphic modal.	Queries available slots via `/api/bookings`; notifies mentor upon submission; mentor approves/reschedules; logs session notes.
EP-04: HR Talent Pool & Discovery	As an HR Recruiter, I want to filter candidates by Builder Score and skill tags so I can shortlist qualified candidates.	Full-text search; domain tag multi-select; score slider; candidate profile drawer displays projects and mentor endorsements.
EP-05: Dynamic Certification	As a Graduate, I want to download an official PDF certificate with an embedded verification QR code.	PDFKit generates a signed certificate; QR code links to public `/verify/:token` resolving student credentials without login.

3 Builder Score Algorithm & Key Metrics

To eliminate subjective manual grading, student competency is represented by the **Builder Score**, calculated through a balanced multi-factor formula:

ALGORITHMIC FORMULA

```
Builder Score = (Events × 0.20) + (Projects × 0.40) + (Mentor Reviews × 0.30) + (Teamwork × 0.10)
```

Jobs are processed in the background via BullMQ. As students attend workshops, complete sprint milestones, and receive mentor reviews, the score updates in near real-time without locking the UI.

Key Performance Indicators (KPIs)

Student Engagement Metrics

Learner KPIs

- **Course Completion Velocity:** Average days to complete a foundational track (<45 days target).
- **Sprint Milestone Throughput:** % of student teams achieving Sprint 3 demo day (>85% target).
- **Mentor Session Utilization:** % of students conducting monthly 1–on-1 advisory sessions (>90% target).

Institutional & Placement Metrics

Enterprise KPIs

- **Campus Placement Rate:** % of qualified students (Builder Score >800 XP) receiving offers (>80% target).
- **Recruiter Conversion Time:** Median days from talent pool shortlisting to technical interview (<7 days).
- **College Multi-Tenant Retention:** Annual institutional renewal rate (>95%).